DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITCS214LA
COURSE TITLE	**:**	Data Structures and Algorithms
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 Unit
LEARNING MODALITY	**:**	Fully Onsite
PRE-REQUISITES	**:**	S-ITCS123 / S-ITCS123LA
CO-REQUISITES	**:**	S-ITCS214
CONSULTATION HOURS	**:**	Saturday: 12:00 – 2:00 COS 100-C Friday: 1:00-3:00 ONLINE
CONTACT INFORMATION	**:**	sperolin@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course provides the concepts of abstract data types of common data structures such as lists, linked lists, stacks, queues, trees and graphs used in solving computer problems. This course also includes the study of time and space complexities of algorithms and the choice of appropriate data structures. The course covers several data structures as tools for programming, best practices in using those data structures with estimated running time and space efficiency, hands-on exercises to develop practical skills in creating impactful programs

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:

CLO1. Understand the importance of data structures and algorithms in application of daily living.
CLO2. Evaluate the running time of algorithms and data structures using big-oh notations.
CLO3. Analyze the features of each data structure and its operations and use it to design algorithms to solve real-world problems.
CLO4. Apply the most appropriate data structures given a specified real world problem.
CLO5. Create codes written in Java for various algorithms and data structures.
CLO6. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN FOR MIDTERM PERIOD:

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Understand data structures & algorithms in application of daily living.	**TLO1.** Distinguish the differences between algorithms and data structures. **TLO2.** Identify the criteria for efficient algorithms.
CLO2. Evaluate the running time of algorithms and data structures using big-oh notations.	**TLO3.** Calculate the running time of codes following specific rules.
CLO3. Analyze the features of each data structure and its operations and use it to design algorithms to solve real-world problems.	**TLO4.** Construct pseudo-language implementations of lists and its operations. **TLO5.** Write algorithms for linked list and its variations. **TLO6.** Understand the concept of a stacks and its operations.
CLO6. Demonstrate 21st century skills in all learning activities.	**TLO7.** Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem- Solving, Use of ICT (not part of topic, but integrated in assessments)

LEARNING OUTCOMES FOR MIDTERM PERIOD

Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy Module Activities Online/ Offline	Technology Enabler	Allotted Hours
Module 0 CLO1 TLO1	**W1** Aug 19 - Aug 24	Gospel Reading, Reflection & Prayer Presentation of the course syllabus **Module 0 – Course Introduction** Introduction of the course professor Presentation of the Enrich Virtual Learning model LMS Formative Activity: Re-familiarization of local and online Java IDEs	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
Module 1 CLO1 TLO2	**W2** Aug 26 – 31	Gospel Reading, Reflection & Prayer **Module 1 – Introduction to Data Structures (Revisit: Arrays and Pointers)** Class Participation Lab Activity Module 1 Enabling Assessment	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
Module 2 TLO3, CLO2, CLO6	**W3** Sep 2 – Sep 7	Gospel Reading, Reflection & Prayer **Module 2 – Introduction to Algorithm Analysis (Java Programs on Big-O functions)** Class Participation Lab Activity Module 2 Enabling Assessment	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3

Module 3 CLO2 TLO3	**W4** Sep 9 – Sep 14	Gospel Reading, Reflection & Prayer **Module 3 – Lists and List Operations** Class Participation Module 3 Enabling Assessment	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
Module 4 TLO5, CLO3, CLO5, CLO6	**W5** Sep 16 – Sep 21	Gospel Reading, Reflection & Prayer **Module 4 – Linked Lists and Linked List Operations** Class Participation Lab Activity for Module 4	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
Module 4 TLO5, CLO3, CLO5, CLO6	**W6** Sep 23 – Sep 28	Gospel Reading, Reflection & Prayer **Module 4 – Linked List Applications** Module 4 Enabling Assessment	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
Module 5 TLO6, CLO3, CLO5, CLO6	**W7** Sep 30 – Oct 5	Gospel Reading, Reflection & Prayer **Module 5: Stacks, Stack Operations and Applications** Class Participation Lab Activity Module 5 Enabling Assessment	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
	W8 Oct 7 - 12	**MIDTERM SUMMATIVE ASSESSMENT** Coverage: Modules 1 to 5	Schoolbook JCreator NetBeans Programiz	3
	W9 Oct 14 – 19	**COMPLETION / CONSULTATION WEEK** *Students will comply with lacking assessments and/or in-person/online consultation with the professor*	DLSU-D Schoolbook	3
			TOTAL	**27**

LEARNING PLAN FOR FINAL PERIOD:

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Apply the most appropriate data structures given a specified real world problem.	**TLO8.** Apply various applications of queue operations to real life scenarios. **TLO9.** Identify the different terminologies used in trees. **TLO10.** Explore different applications of binary trees. **TLO11.** Discover the varied ways of graph representations and applications. **TLO12.** Determine the steps in creating recursive routines.
CLO5. Create codes written in Java for various algorithms and data structures.	**TLO13.** Use and apply Java structure and elements to implement the algorithms and data structures covered in this course.
CLO6. Demonstrate 21st century skills inall learning activities.	**TLO7.** Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

LEARNING OUTCOMES FOR FINAL PERIOD

Module Title and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategy	Technology Enabler	Allotted Hours
Module 6 TLO8, CLO4, CLO5, CLO6	**W10** Oct 21 – 26	Gospel Reading, Reflection & Prayer **Module 6: Queues and Queue Operations** Module 6 Enabling Assessment	DLSU-D Schoolbook Newsfeed	3
	W11 Oct 28 – Nov 2	**Independent Learning Week**	DLSU-D Schoolbook Newsfeed Self-care Hub	3
Module 7 TLO9, CLO4, CLO5, CLO6	**W12** Nov 4 – Nov 9	Gospel Reading, Reflection & Prayer **Module 7: Trees and Tree Operations** Module 7 Enabling Assessment	Schoolbook, JCreator NetBeans Programiz Notepad	3
Module 8 TLO10, CLO4, CLO5, CLO6	**W13** Nov 11 - Nov 16	Gospel Reading, Reflection & Prayer **Module 8: Binary Trees and its Operations** Class Participation Lab Activity	Schoolbook, JCreator NetBeans Programiz GitHub	3
Module 8 TLO10, CLO4,	**W14** Nov 18-Nov 23	Gospel Reading, Reflection & Prayer **Module 8: Trees and Binary Trees' Applications**	Schoolbook, JCreator NetBeans	3

CLO5, CLO6		Module 8 Enabling Assessment	Programiz	
Module 9 TLO11, CLO4, CLO5, CLO6	**W15** Nov 25 – Nov 30	Gospel Reading, Reflection & Prayer **Module 9: Graphs and Graph Operations** Class Participation Lab Activity for Module 9	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
Module 9 TLO11, CLO4, CLO5, CLO6	**W16** Dec 2 – Dec 7	Gospel Reading, Reflection & Prayer **Module 9: Graphs' Applications** Module 9 Enabling Assessment	Schoolbook, JCreator NetBeans Programiz GitHub Stackoverflow	3
	W17 Dec 9-14	**FINALS SUMMATIVE ASSESSMENT** Coverage: Modules 6 to 9	Schoolbook JCreator NetBeans Programiz	3
	W18 Dec 16- Dec 18	**COMPLETION / CONSULTATION WEEK** *Students will comply with lacking assessments and/or in-person/online consultation with the professor*	DLSU-D Schoolbook	3
			TOTAL	**27**

GRADING SYSTEM

Midterm

Enabling Assessments (EA)	-	40%
Summative Assessments (SA)	-	60%
Total		100%

Finals

Enabling Assessments (EA)	-	40%
Summative Assessments (SA)	-	60%
Total		100%

Overall Final Grade = [(Midterm grade) + (Final term grade)] / 2

Note: Class participation (CP) refers to your active engagement in class from day 1 to the last day of each term (midterm or final). CP includes, but not limited to, the following: actual attendance, observance of netiquettes (synchronous) and etiquettes, promptness in submitting tasks/assessments, relevant questions asked, participation in collaborative tasks, etc.

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
    a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
    a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

Call number or e-provider	Reference Material
QA 76.9.S88 .H191 2018	Mehta, D. P., & Sahni, S. (2018). Handbook of data structures and applications (Second edition.). CRC Press/Taylor & Francis Group.
QA 76.64 .D152 2018	Dale, N., Joyce, D. T., & Weems, C. (2018). Object-oriented data structures using Java (Fourth edition.). Jones & Bartlett Learning.

ONLINE REFERENCES

Call number or e-provider	Reference Material
e-Book	John Bullinaria 2019. Data Structures and Algorithms - School of Computer Science. Universityof Birmingham, UK. [accessed 2024 August 10] https://www.cs.bham.ac.uk.
e-Book	Goodrich, M. T. Data Structures and Algorithms in Java. Department of Computer Science University of California, Irvine. John Wiley & Sons, Inc. Retrieved from https://enos.itcollege.ee/~jpoial/algorithms/GT
YouTube Channel	Computer Education or all. 2018. Data Structures and Algorithms Complete Tutorial Computer Education for All [accessed 2024 August 15] https://www.youtube.com/watch?v=YWnBbNj_G-U
YouTube Channel	Tech Course PH. 2020. Data Structures and Algorithms Lecture Series. [accessed 2024 August 15] https://www.youtube.com/watch?v=CNhj4GZ_7iE

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